VELOOP REWARDS - Analytics Change Management

Data Analyst Internship, Task 01 | Work Package Q

Authenticity notice: This is a **Proposed** process. No simulated or real VELOOP statistics appear here. Dates in the log are entered when each change is approved.

1. Summary

- **VELOOP will keep changing,** so analytics needs a light way to change with it without breaking history or trust. This proposal gives a nine-step process, a Breaking / Compatible / Cosmetic classification, an impact matrix and a Metric / Event Change Log.
- **All 7 triggers in the brief are covered** (feature, event, reward mechanic, withdrawal method, field rename, KPI definition, dashboard), plus two added ones: a new feature launch and a provider change.
- **The change log already holds seven real entries** from aligning this document set and from the dataset work (for example, 'Completion Rate' split into Task and Withdrawal Completion Rate). Dates are left to enter on approval.

2. Principles

- Nothing changes silently: every change to an event, field, KPI, dashboard or rule has a log entry, an owner and a version.
- Classify before acting: a change is Breaking, Compatible or Cosmetic, and the class decides the handling.
- Find the impact first: use the impact matrix to see which documents, KPIs and dashboards a change touches.
- Old and new must stay explainable: history is either kept under the old definition or restated and labelled, never mixed unlabelled.
- Approve by ownership: the data owner approves; the Data Analyst assesses impact; Engineering implements.
- Keep it light: a shared change log and short change requests are enough at the start.

3. Change process

Step	Action	Who	Output
1	Propose the change with a short change request	Anyone	Change request (template below)
2	Assess impact using the impact matrix and the dependency lists in each document	Data Analyst	List of affected events, fields, KPIs, dashboards, reports and rules
3	Classify the change as Breaking, Compatible or Cosmetic	Data Analyst	Class and version impact
4	Approve, reject or defer	Data owner (Product); Finance consulted for money KPIs	Decision recorded
5	Implement in order: reference data and events, then pipeline and rules, then KPIs, then dashboards	Engineering, Data Analyst	Changed artifacts
6	Test with the acceptance tests and quality rules affected	Data Analyst, Engineering	Test result
7	Release with a new version and tell affected users	Data Analyst	Release note
8	Update the change log and every affected document	Data Analyst	Log entry, updated documents
9	Review after release: did KPIs and dashboards behave as expected?	Data Analyst	Follow-up actions

4. Change classes

Class	Meaning	Examples	Handling	Version effect
Breaking	Existing reports or consumers could give different or wrong results	Rename or remove an event or field; change what a status or unit means; change a KPI formula; change the activation rule; change how an amount is signed	Impact assessment required; announce before release; run old and new side by side for an overlap period or restate history with a label (lengths to be agreed)	Major version
Compatible	Adds something without changing existing meaning	New event; new optional property; new allowed value that consumers tolerate; new KPI; new dashboard visual	Impact check; update documents; no restatement	Minor version
Cosmetic	Wording or layout only; no effect on data or numbers	Better description; dashboard layout or colour; typo fix	Log entry only	Patch (no data effect)

5. Playbook for each trigger

Trigger	What Usually Changes	Must Be Reviewed or Updated	Class (typical)	Special Handling	Approval
A feature changes	Events, states and fields for that feature	Event Tracking Plan, dictionary, KPI formulas that use the feature, dashboards, acceptance tests	Compatible or Breaking	Confirm which events still make sense; retire events of removed features	Product
An event changes (new, modified, removed, renamed)	Name, trigger, required properties or meaning	Event Tracking Plan, dictionary, quality rules (DQ020, DQ022), developer handoff, KPI dependencies, pipeline validation	Compatible if added; Breaking if renamed, removed or redefined	Renaming is a new event plus a deprecation of the old; keep emitting both for the overlap period; bump schema_version	Product (owner); Engineering confirms
A reward mechanic changes	Amounts, units, validation, pending or reversal rules	Reward and wallet fields, reward units list, KPIs R01 to R07 and W02, dashboards 4 and 5, reconciliation rules DQ030 and DQ034	Breaking if meaning or unit changes	Record the date the mechanic applies from; compare periods only within one mechanic or label the change on charts	Finance and Product
A withdrawal method changes	Allowed method values, states, processing steps, reason categories	Approved lists, withdrawal fields and states, KPIs D01 to D10, dashboard 6, transition rules DQ026	Compatible if a method is added; Breaking if states or meaning change	Add the value to the approved list before release so events are not quarantined; mark the date on trend charts	Operations and Finance
A field is renamed	Column name in operational and analytics tables	Dictionary, logical data model, event properties, payload spec, KPI required fields, quality rules, pipeline jobs	Breaking	Prefer adding the new field and deprecating the old; map names in the transformation layer until consumers move	Data owner; Engineering
A KPI definition changes	Formula, filter, period or denominator	KPI Catalogue, dashboard visuals, pipeline views, any report quoting it	Breaking	New KPI version; label dashboards 'definition vN applies from the change date'; keep old values under the old version or restate history, with a label	Product; Finance for money KPIs
A dashboard changes	Layout, filters, visuals	Dashboard blueprint, KPI mapping, access rules	Cosmetic or Compatible; Breaking if a visual's calculation changes	Any change to a calculation is a KPI change, not a dashboard change	Product
A new feature launches (added)	New events, fields, KPIs	Registers, event plan, dictionary, logical model, KPI Catalogue, quality rules, dashboards, handoff	Compatible	Move assumptions from To Be Verified to Confirmed only with evidence	Product
A third-party provider changes (added)	Callbacks, statuses, reference ids	Provider fields, failure categories, reliability KPI T02, pipeline handling	Compatible or Breaking	Keep VELOOP status and provider status separate; reconcile with the provider's report	Engineering and Product

6. Impact matrix

U = update, R = review, - = not affected. Numbers refer to the document set.

Trigger	S1 Registers	08 Event Plan	09 Dictionary	10 Data Model	11 KPI Catalogue	S2 Quality Rules	14 Dashboards	15 Dev Handoff	16 Pipeline	Event schema_version
A feature changes	U	U	R	R	R	R	R	U	-	-
An event changes	R	U	U	R	R	U	R	U	R	U
A reward mechanic changes	U	R	U	R	U	R	R	R	-	R
A withdrawal method changes	U	R	U	-	R	U	R	R	-	-
A field is renamed	-	U	U	U	U	U	R	U	U	U
A KPI definition changes	-	R	R	-	U	R	U	R	R	-
A dashboard changes	-	-	-	-	R	-	U	-	R	-
A new feature launches	U	U	U	U	U	U	U	U	R	-
A third-party provider changes	U	R	R	-	R	R	-	R	R	-

7. Versioning

Object	Version Scheme	Where Kept	Illustrative Example
Event definition	Event schema_version in every payload carries the major version only; minor changes are recorded in the catalogue	Event Tracking Plan; every event payload	schema_version = 2 after a breaking change
KPI	KPI ID with a version: major.minor	KPI Catalogue and dashboard tooltip	E05 v1.0, then E05 v2.0 if the formula changes
Data dictionary and logical model	One document version, major.minor	Document header and change log	Dictionary v1.1 after a field is added
Quality rule	Rule ID with a revision	Quality framework	DQ013 rev 2
Dashboard	Dashboard version, major.minor	Dashboard blueprint	Executive Overview v1.2
Approved lists	Version of the reference table	Reference data	Withdrawal methods list v3

8. Handling history

- Keep or restate: when a KPI or event definition changes, either keep history under the old version and start the new one from the change date, or recalculate history under the new definition. Either way the dashboard shows a marker and a label.
- Overlap period: for a breaking change, old and new run side by side for an overlap period (length to be agreed) so users can compare.
- Deprecation: a retired event or field is marked Deprecated, still documented, and removed only after consumers have moved and the owner approves.
- Raw data is never edited to fit a new definition; new logic is applied in the transformation layer, so the old result can be reproduced.

9. Responsibilities (RACI)

Activity	Anyone	Data Analyst	Engineering	Product (data owner)	Finance / Operations
Raise a change request	R	I	I	I	I
Assess impact and classify	C	A/R	C	C	C
Approve or reject	I	C	C	A/R	C
Implement events, fields and reference data	-	C	A/R	I	I
Implement KPI, rule and dashboard changes	-	A/R	C	I	I
Test and verify	-	A/R	R	I	I
Release note and notify users	-	A/R	C	C	I
Update the change log and documents	-	A/R	C	I	I

10. Metric / Event Change Log

Template columns follow the brief: Date, Change, Old Definition, New Definition, Reason, Affected KPIs, Affected Dashboards, Affected Reports, Owner, Version. Extra columns (Change ID, Type, Class, Approved By, Status) help tracking. The full log is in S3_Analytics_Change_Management.xlsx.

ID	Change	Old Definition	New Definition	Reason	Affected KPIs	Affected Dashboards	Owner	Version	Class
CL-001	Event Tracking Plan KPI Dependency regenerated from the KPI Catalogue	Preliminary KPI names; 'Completion Rate' used for both tasks and withdrawals	KPI names taken from the KPI Catalogue; 'Completion Rate' split into 'Task Completion Rate' and 'Withdrawal Completion Rate'	Remove ambiguity between two different rates	E05, D07	1 (Executive Overview), 3 (Engagement), 6 (Withdrawals)	Data Analyst	Event Plan 1.1	Compatible
CL-002	Data Coverage Matrix aligned with the Event Tracking Plan	36 rows; some event names marked 'to be agreed'	42 rows; adds account_status_changed, reward_rejected, mining_started, mining_claimed, feature_opened, withdrawal_cancelled; names finalised	Documents must agree on events, sources, statuses and priorities	None	None	Data Analyst	Coverage Matrix 1.1	Compatible
CL-003	Task-to-reward link designated	Task_Activity.reward_id and Rewards.task_activity_id both listed with no authority stated	Rewards.task_activity_id is authoritative; Task_Activity.reward_id is a convenience copy checked by rule DQ034	Two copies of one link must not disagree	None directly	None	Data Analyst	Data Model 1.0 note	Compatible
CL-004	Event name suffixes removed after catalogue finalised	wallet_exchange_completed, support_request_created and provider_callback_received marked 'name to be agreed'	Names used as written; status stays To Be Verified	Consistency across documents	W05, S01, S02, T02	5 (Wallet), 3 (Engagement), 1 (Executive)	Data Analyst	Event Plan 1.1	Compatible
CL-005	Assumptions A017 and A043 confirmed and A046 to A050 added after the supplied dataset was received	A017 and A043 To Be Verified; 45 assumptions	A017 and A043 Confirmed from the supplied file; 50 assumptions	Dataset evidence resolved two unknowns and raised five new ones	None	None	Data Analyst	Register 1.1	Compatible
CL-006	Readiness raised from R1 to R2 for the data dictionary, KPI definitions and data quality	All ten areas at R1 (designed on paper)	Three areas at R2 (checked against the supplied Users dataset); seven remain at R1	The Users data was profiled, audited and used for the KPI test	All (KPI test)	None	Data Analyst	Readiness 1.1	Compatible
CL-007	Quality rules DQ040 to DQ042 proposed by the Data Quality Audit	39 rules in the Data Quality Control Framework	39 rules plus 3 proposed (derived-field consistency, counter consistency, consent consistency); DQ013 and DQ039 extensions proposed	The audit found problems that existing rules cannot detect	None	None	Data Analyst	Quality Framework 1.1 (proposed)	Compatible

11. Change request template

Field	What to Enter
Change request ID	CR-###
Title	Short name of the change
Raised by and date	Name and date
Trigger type	Feature, event, reward mechanic, withdrawal method, field rename, KPI, dashboard, new feature or provider
Description	What changes, in plain words
Reason	Why the change is needed
Old definition	Exact current definition
New definition	Exact proposed definition
Affected artifacts	Use the impact matrix: documents, events, fields, KPIs, dashboards, reports, rules
Class	Breaking, Compatible or Cosmetic
History handling	Keep old, restate, or overlap period
Approver and decision	Data owner and result
Version after release	For example 1.2 or 2.0
Test evidence	Acceptance tests and quality rules run

12. Document versions

Document	Version	Status	Alignment Notes
Source and Verification Register	1.0	Proposed	Baseline; superseded for assumptions by the Assumption Register
Product Data Audit	1.1	Proposed	Text updated to match the 42-row matrix
Data Coverage Matrix	1.1	Proposed	Aligned with the Event Tracking Plan (CL-002)
User Analytics Specification	1.0	Proposed	Baseline
Earning and Reward Analytics Specification	1.0	Proposed	Baseline
Wallet and Withdrawal Analytics Specification	1.0	Proposed	Baseline
Referral and Engagement Analytics	1.0	Proposed	Baseline
Event Tracking Plan	1.1	Proposed	KPI Dependency regenerated from the KPI Catalogue (CL-001, CL-004)
Data Dictionary	1.0	Proposed	Baseline
Logical Data Model and ER Diagram	1.0	Proposed	Link authority noted (CL-003)
KPI Catalogue	1.0	Proposed	Baseline
Data Quality Control Framework	1.0	Proposed	Baseline; DQ040 to DQ042 proposed (CL-007)
Dashboard Blueprint	1.0	Proposed	Baseline
Developer Analytics Handoff	1.0	Proposed	Baseline
Data Pipeline Proposal	1.0	Proposed	Baseline
Privacy and Governance Proposal	1.0	Proposed	Baseline
Simulated Data Profiling	1.0	Proposed	Baseline; describes the supplied simulated Users dataset only
Data Quality Audit	1.0	Proposed	Baseline; describes the supplied simulated Users dataset only
Assumption and Verification Register	1.1	Proposed	A017 and A043 confirmed; A046 to A050 added (CL-005)
Analytics Readiness Assessment	1.1	Proposed	Three areas raised to R2 (CL-006)

Document	Version	Status	Alignment Notes
Implementation Backlog	1.1	Proposed	Items added for datasets, derived fields and data minimisation
Executive Summary and Final Consolidated Report	1.1	Proposed	Updated for the dataset results

13. Open questions

- How long should the overlap period be for breaking changes? (No length is set here.)
- How much notice should dashboard users get before a breaking change?
- Who approves money-related KPI changes besides Product (Finance, Operations)?
- Where will the change log live: a shared spreadsheet at first, with the documents under version control?
- How are dashboard users told about changes (banner, message)?

Next: Work Package R (Analytics Gap Assessment) and S (Assumption and Verification Register).